

The methanol extract of *Euonymus laxiflorus*, *Rubia lanceolata* and *Gardenia jasminoides* inhibits xanthine oxidase and reduce serum uric acid level in rats



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ABSTRACT

Chinese herbal medicinal plants, *Euonymus laxiflorus* (EL), *Rubia lanceolata* (RL) and *Gardenia jasminoides* (GJ), have been used widely to treat arthritis and gout in Taiwan for decades. To understand the beneficial effects of these three plants, their xanthine oxidase (XO) inhibitory activity *in vitro* and hypouricaemic activity *in vivo* were investigated. Our results suggested that methanol extracts were better than water extracts for inhibition of XO activity and 2,2-diphenyl-1-picrylhydrazyl (DPPH) radical scavenging activity, except the water extract of GJ, which exhibited the strongest radical scavenging effect. In animal study, the serum urate level was significantly decreased after oral administration of higher dose (0.39 g/kg) methanol extract of the mixture of three plants (ERG). In addition, methanol extract of ERG reduced the pain reaction time in the second phase of formalin induced pain. The results provide useful information on the pharmacological activities of these plants for the potential in treating hyperuricemia.

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1. Introduction

Gout is a common metabolic disorder with a worldwide distribution and continues to be a major health problem. Hyperuricemia, which is associated with gout, results from the overproduction or underexcretion of uric acid in the body. The deposition of urate crystals in the joints and kidneys causes inflammation as well as gouty arthritis and uric acid nephrolithiasis (Kramer and Curhan, 2002; Tomita et al., 2000). More importantly, uric acid is

not only related to an increased risk of gout, but also to cardiovascular disorder, nephrolithiasis and diabetes (Colvine et al., 2008; Corrado et al., 2006). It is widely accepted that the control of the uric acid may be considered in the prevention and treatment of these diseases.

Xanthine oxidase (XO), which is present in significant concentration only in liver and intestine, oxidizes hypoxanthine and xanthine to uric acid in the purine catabolic pathway. Treatment and prevention of gout entail to reduce serum uric acid concentration in the body. XO inhibitors are available to block the synthesis of uric acid in the body, as well as anti-inflammatory agents to relieve the symptoms of the disease (Sarkozi, 1996). Allopurinol is the only inhibitor of the enzyme in clinical use, but its use can result in a number of adverse side effects, such as allergic and hypersensitivity reactions, nephropathy, and enhancement of 6-mercaptopurine toxicity (Fels and Sundry, 2008). Thus, the development of new hypouricemic agents of greater effectiveness and safety is highly desirable.

The ability of XO to produce reactive oxygen species (ROS) has led to widespread interest in the enzyme as an initiator of tissue

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damage in a range of pathological states (Bonomini et al., 2008). Hydrogen peroxide (H_2O_2) and superoxide anion radicals ($\text{O}_2^{\cdot-}$) can interact in the presence of certain transition metal ions to yield a highly reactive oxidizing species, the hydroxyl radical ($\cdot\text{OH}$). Continuous exposure to chemicals and contaminants may lead to an increase in the amount of free radicals in the body. Free radicals have been regarded as a fundamental cause of different kinds of diseases, including aging, coronary heart disease, inflammation, stroke, diabetes mellitus, rheumatic disease, liver disorders, renal failure and cancer (Bulkley, 1983; Cheng et al., 2003; Dormandy, 1983).

The use of natural products is gaining renewed interest in connection with the treatment of some diseases. Scientists have turned to explore the potent XO inhibitor from a wide variety of traditional folk medicines. To identify potential XO inhibitory agents from natural sources, we have tested 3 selected medicinal plants, which are used by the indigenous people in Taiwan for the treatment of gout. *Gardenia jasminoides* is widely used in several Asian countries as a natural colorant, and has been used in Chinese traditional medicine for its cholagogue, sedative, diuretic, antiphlogistic, homeostatic, and antipyretic effects (Jung et al., 2008; Luo et al., 2007). *Rubia lanceolata* is endemic in Taiwan, and is a perennial climbing herb that grows throughout the island and distributes in forests at middle to high altitudes (Kuo et al., 1995). The root of *Rubia* species is an important Chinese folk medicine which possesses circulation-promoting, expectorant, against cough properties and antitumor activities. The aim of the present study was to investigate the potential of selected medicinal plants to inhibit XO and to act as an antioxidant and free radical scavenging material and examine for their phenolic contents *in vitro*. Moreover, the extract was investigated for its effects on elimination of uric acid in urine *in vivo*. Screening of the extracts for the XO inhibitory activity followed by their potential to reduce the serum urate level may play a crucial role in identifying a potent chemical entity for treating gout and related inflammatory disorders.

2. Materials and methods

2.1. Chemicals and reagents

Xanthine, XO, DPPH (2,2-diphenyl-1-picrylhydrazyl), vitamin E (α -tocopherol), Folin–Ciocalteu reagent, chlorogenic acid, gallic acid, allopurinol, and oxonate (oxonic acid potassium salt) were purchased from Sigma Chemical Co. (St. Louis, MO, USA). DMSO (dimethyl sulfoxide) and methanol were purchased from Tedia Co. (St. Fairfield, OH, USA). All other chemicals were of analytical reagent grade. Allopurinol was dissolved initially in DMSO (at 0.25% v/v) followed by dilution in PBS solution (phosphate-buffer saline; pH 7.5) to desired concentrations before the use.

2.2. Preparation of *Euonymus laxiflorus*, *Rubia lanceolata* and *Gardenia jasminoides* extracts

Euonymus laxiflorus (EL), *R. lanceolata* (RL) and *G. jasminoides* (GJ) were collected from local Chinese herb markets during the month of April, and identified by one of authors, Prof. Jih-Jung Chen. The voucher specimens have been deposited at the herbarium of Graduate Institute of Pharmaceutical Technology, Tajen University. The voucher specimen number of *E. laxiflorus* is 74279, *R. lanceolata* is 72491, and *G. jasminoides* is 054093, respectively. After being dried and cut up, each and mix of three plants (ERG) were extracted by macerating and refluxing them in methanol and water for 3 h. The ERG is a mixture of equal amounts of three medicinal plants. The extracts respectively are EL/ H_2O , RL/ H_2O , GJ/ H_2O , ERG/ H_2O , EL/MeOH, RL/MeOH, GJ/MeOH and ERG/MeOH. Those extracts were filtered for removal of peel particles. After solvent removal in a rotary evaporator (Eyelan-N, Tokyo Rikakikai Co. Ltd., Japan), the extracts of methanol and water obtained were assessed for their biological activities.

2.3. Analysis of total phenolic content

The total phenolic content of the aqueous and methanol extracts of EL, RL, GJ and ERG were analyzed according to the Folin–Ciocalteu method (Cliffe et al., 1994). The reaction solution consisting of 0.2 ml of the extracts, 1 ml of 7.5% Na_2CO_3 reagent and 1 ml of the Folin–Ciocalteu stock reagent (10X) were well mixed and incubated at room temperature in dark for 30 min. The standards were prepared

in the same way. The mixture absorbance was measured spectrophotometrically at wavelength 765 nm. The total phenol content was expressed in milligrams of gallic acid or chlorogenic acid equivalents per gram of those extracts.

2.4. DPPH free radical-scavenging assay

Quantitative measurement of radical scavenging properties was carried out in a 96-well plate assay. The free radical-scavenging activity of the aqueous and methanol extracts of EL, RL, GJ and ERG on the DPPH radical was assessed using the method described previously (Hu et al., 2004), with some modifications. A stock solution (1 mg/ml) of each extract was prepared and diluted with methanol to various concentrations. An aliquot of 50 μL of each dilution was transferred into a 96-well microplate (NUNC, Roskilde, Denmark). A working solution of DPPH (250 μM) in methanol was freshly prepared and then an aliquot of 150 μL was added to each well. After incubation for 30 min, the quenching at an absorbance of 517 nm was measured on an ELISA reader (ThermoLabsystems, Cheshire, UK). Each dilution was performed at least in triplicate. Free radical-scavenging activities of test samples and the positive control (Vit. E) were expressed in terms of IC_{50} values, which is the concentration of a sample required to decrease the absorbance at 517 nm by 50% compared to the control response.

2.5. Assay of XO inhibitory activity

The inhibitory effect on XO was assayed spectrophotometrically at 295 nm under aerobic condition (Kong et al., 2000; Nguyen et al., 2004; Unno et al., 2004) with modification by using 96-well plates. The reaction solution consisting of 50 μL test solution dissolved in nanopure water or DMSO, 35 μL PBS (50 mM, pH 7.5), and 30 μL XO (0.1 units/mL in 50 mM PBS, pH 7.5) was prepared immediately before use. DMSO was used for the samples not dissolvable in distilled water. One unit of XO is defined as the amount of enzyme required to produce 1 μmol of uric acid/min at 25 °C. The reaction solution was well mixed and incubated at 25 °C for 15 min. The reaction was initiated by the addition of 60 μL xanthine solution (150 mM) in the same buffer (substrate solution), and incubated for 30 min at the same temperature. The reaction was stopped by adding 25 μL of 1 N HCl, and the absorbance at 290 nm was measured with a Universal Microplate Spectrophotometer SpectraMax190 (Molecular Devices Corporation, Sunnyvale, California, USA). The blank and positive control groups were prepared in the same way, but each group was added 50 μL PBS and allopurinol without test solution.

The XO inhibitory activity was expressed as the percentage inhibition of XO calculated as % Inhibition = $(1 - B/A)$, where A (ΔA_{blank} with enzyme – ΔA_{blank} without enzyme) and B (ΔA_{test} with enzyme – ΔA_{test} without enzyme) are the activities of the enzyme without and with test solution. The crude extracts were dissolved initially in DMSO followed by dilution with PBS, and the final concentration of DMSO was less than 0.25%.

2.6. In vivo studies

The methanol extract of the mix of *E. laxiflorus*, *R. lanceolata* and *G. jasminoides* (ERG/MeOH) produced significantly higher *in vitro* XO inhibitory activity than those of water extract. Thus we selected ERG/MeOH for the screening of uric acid clearance via hyperuricemic rat model *in vivo*.

2.6.1. Preparation of the hyperuricemic rat model

Wistar rats (10 weeks old, 230–280 g) were purchased from the National Laboratory Animal Breeding and Research Center (Taipei, Taiwan) following China Medical University Institutional Animals Ethics Committee clearance (99-151-N). The rats were housed in stainless steel wire-bottomed cages and acclimated under laboratory conditions (19–23 °C, humidity of 60%, and 12 h light/12 h dark cycle) for at least one week before each study. Rats were fasted before administration of test substances. The hyperuricemic rat model was prepared by intraperitoneal (i.p.) administration of oxonate (250 mg/kg single dose) according to the method described previously (Yonetani et al., 1980). Oxonate for administration was suspended in 0.5% carboxymethylcellulose (CMC) aqueous solution.

2.6.2. Hypouricaemic activity study

Hyperuricemic rats (i.p. oxonate 250 mg/kg) were assigned to this experiment, six groups of rats ($n = 8$ –10 per group): Group 1 was control group of normal rat. This group was only administered with vehicle 0.5% CMC solution. In group 2, the animals only received 0.39 g/kg ERG/MeOH. Group 3 was intraperitoneally administered with potassium oxonate (250 mg/kg) and served as hyperuricemic control. Groups 4 and 5 received the oral treatment with 0.13 g/kg and 0.39 g/kg ERG/MeOH in the hyperuricemic rats, respectively. Group 6 received oral treatment with allopurinol 10 mg/kg as positive control. Each sample was dissolved in 0.5% CMC solution. Blood from orbital vascular plexus was collected at the different times (0, 60, 90, 120, 180, and 270 min). The blood was allowed to clot and serum separated. The serum was separated immediately after the blood collection. The serum urate level was determined using the method described previously (Nguyen et al., 2005; Basah et al., 2011).

2.6.3. Acetic acid-induced writhing

This test was performed as previously described (Koster et al., 1959). Groups of 10 mice were fasted overnight prior to the beginning of the experiment, while free access to water was given. Acetic acid (0.6% v/v) was administered i.p. in a volume of 10 mL per 1 kg of animal. Mice were then placed in boxes and the number of writhes, a response consisting of contraction of the abdominal wall and pelvic rotation followed by hind limb extension, was counted during continuous observation for 20 min, starting after 5 min of the acetic acid injection. ERG/MeOH (0.13 g/kg and 0.39 g/kg), indomethacin (20 mg/kg) and equivalent volumes of vehicle were administered 60 min before the acetic acid injection. Indomethacin, a well-known peripheral analgesic drug, was used as positive control. Antinociceptive activity was expressed as percentage of inhibition of the usual number of writhings observed in control animals.

2.6.4. Formalin-induced nociception

The procedure reported here was already described (Dubuisson and Dennis, 1977; Hunskaar and Hole, 1987). Animals received 20 μ L of formalin solution (5% formaldehyde in saline) administered to mice via intraplantar route and immediately the licking time was registered for 5 min to determine the first phase (i.e. neurogenic). 15 min after the beginning of the experiment, the licking time was registered for another 15 min, corresponding to the second phase (inflammatory). The responses caused by first and second phases of the formalin test were recorded for 30 min after ERG/MeOH (0.13 g/kg and 0.39 g/kg), indomethacin (20 mg/kg) and vehicle administration.

2.6.5. Acute toxicity studies

ICR mice of either sex (6 week old, 20–25 g, $n = 10$) maintained under standard laboratory conditions were used. A total of five animals for either sex was used which received a single oral dose (5 g/kg) of ERG/MeOH. Animals were kept overnight fasting prior to drug administration. After the administration of the extract, food was withheld for further 3–4 h. Animals were observed individually at least once during the first 24 h (with special attention to first 4 h) and daily thereafter for a period of 14 days. Mortality, if any, was determined over a period of two weeks.

2.7. Statistical analysis

The data were analyzed using SigmaPlot 12.0 (Systat Software Inc., Chicago, USA). Values were represented as mean $\pm$ S.D. For multiple comparisons, one way ANOVA was used followed by Scheffe test and/or Dunnett test. * indicates the P values are significantly different from the control (* $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$).

3. Results

3.1. Yields of EL, RL, GJ and ERG extracts obtained by methanol and water extraction methods

The yields of EL, RL, GJ and ERG extracts obtained by water were 6.0%, 12.3%, 4.9% and 11.6%, and by methanol were 4.5%, 32.7%, 6.6% and 3.5%, respectively.

3.2. Total phenolic contents of EL, RL, GJ and ERG extracts

The total phenolic content was expressed in milligrams of gallic acid or chlorogenic acid equivalents per gram of EL, RL, GJ and ERG

extracts. The values of total phenols in EL, RL, GJ and ERG extracts are shown in Table 1. The total phenolic contents range from 105.11 ± 7.46 to 267.18 ± 10.76 mg equivalent gallic acid/g and from 105.07 ± 7.31 to 263.96 ± 10.55 mg equivalent chlorogenic acid/g, respectively. The highest level of phenols was recorded in the GJ/H₂O extract.

3.3. Free radical scavenging activity

The DPPH method has been widely accepted as a measure for free radical-scavenging activities of antioxidants (Hu et al., 2004; Jin and Chen, 1998; Kim et al., 2002; Leong and Shui, 2002). The percentages of DPPH remaining in the presence of EL, RL, GJ and ERG water extracts at concentrations 1000 μ g/mL were $85.91 \pm 0.23\%$, $78.86 \pm 0.19\%$, $87.49 \pm 0.05\%$, $87.90 \pm 0.28\%$ and methanol extracts were $89.0 \pm 0.16\%$, $70.58 \pm 1.22\%$, $90.9 \pm 0.19\%$, $88.02 \pm 0.14\%$, respectively. The antioxidant activity, the IC₅₀, which is the concentration of sample required to decrease the absorbance at 490 nm by 50% for extracts are shown in Table 2. The IC₅₀ values of EL, RL, GJ and ERG extracts obtained by water were 379.6 ± 8.7 , 591.2 ± 3.0 , 74.6 ± 3.6 , 356.4 ± 15.1 μ g/mL and methanol were 115.6 ± 2.9 , 538.2 ± 14.8 , 123.1 ± 1.0 , 187.4 ± 16.9 μ g/mL, respectively. The order of free radical-scavenging activity was vitamin E > GJ/H₂O > EL/MeOH > GJ/MeOH > ERG/MeOH > ERG/H₂O > EL/H₂O > RL/MeOH > RL/H₂O. The GJ/H₂O extract exhibited a potent scavenging effect comparable to Vitamin E, a well known radical scavenging compound.

3.4. Xanthine oxidase inhibitory activity in vitro

The XO inhibitory effects of the methanol and water extracts isolated from the EL, RL, GJ and ERG were compared with allopurinol, which is clinically used as a drug for the XO inhibitor. As shown in Fig. 1(A) and (B), all extracts exhibited inhibitory activity against the XO-induced superoxide formation at concentrations 1000 μ g/mL. The order of XO inhibition effect was allopurinol (25 μ g/mL) > RL/MeOH > ERG/MeOH > RL/H₂O > GJ/H₂O > EL/H₂O > EL/MeOH > ERG/H₂O > GJ/MeOH. Methanol extracts of *G. jasminoides* did not inhibit XO inhibitory activity. Moreover, Fig. 1(C) shows that the strongest XO inhibitor, RL/MeOH, exhibited inhibitory activity against the XO-induced superoxide formation in a concentration-dependent manner. The XO inhibitory activity of methanol extract of ERG is better than water extract.

3.5. Hypouricaemic activity in vivo

The methanol extract of the mixture of three plants (ERG) are used by the indigenous people in Taiwan for the treatment of gout,

Table 1

Total phenols contents of the different extracts. Concentration of individual and total phenols determined the different extracts that methanol and water extracted from *Euonymus laxiflorus* (EL), *Rubia lanceolata* (RL) and *Gardenia jasminoides* (GJ) by gallic acid (mg eq./g of dry weight) and chlorogenic acid (mg eq./g of dry weight).

The different extracts	Gallic acid (mg/g)	Chlorogenic acid (mg/g)
EL/MeOH	215.38 ± 8.90	213.18 ± 8.73
EL/H ₂ O	105.11 ± 7.46	105.07 ± 7.31
RL/MeOH	117.98 ± 5.17	117.69 ± 5.07
RL/H ₂ O	124.98 ± 6.12	124.55 ± 6.00
GJ/MeOH	240.05 ± 4.46	237.36 ± 4.37
GJ/H ₂ O	267.18 ± 10.76	263.96 ± 10.55
ERG/MeOH	206.25 ± 3.95	204.22 ± 3.87
ERG/H ₂ O	142.11 ± 5.40	141.35 ± 5.30

Results were expressed as average (mean) concentrations $\pm$ S.D. of triplicate.

Table 2

Antioxidant activities of the different extracts were using the DPPH free radical-scavenging assay. Results were expressed as average (mean) concentrations $\pm$ S.D. of triplicate.

Sample	IC ₅₀ (μ g/mL) ^a
Vitamin E	22.0 ± 0.5
EL/MeOH	115.6 ± 2.9
EL/H ₂ O	379.6 ± 8.7
RL/MeOH	538.2 ± 14.8
RL/H ₂ O	591.2 ± 3.0
GJ/MeOH	123.1 ± 1.0
GJ/H ₂ O	74.6 ± 3.6
ERG/MeOH	187.4 ± 16.9
ERG/H ₂ O	356.4 ± 15.1

^a The antioxidant activity was evaluated as the concentration of the test sample required to decrease the absorbance at 490 nm by 50% in comparison to the control response.

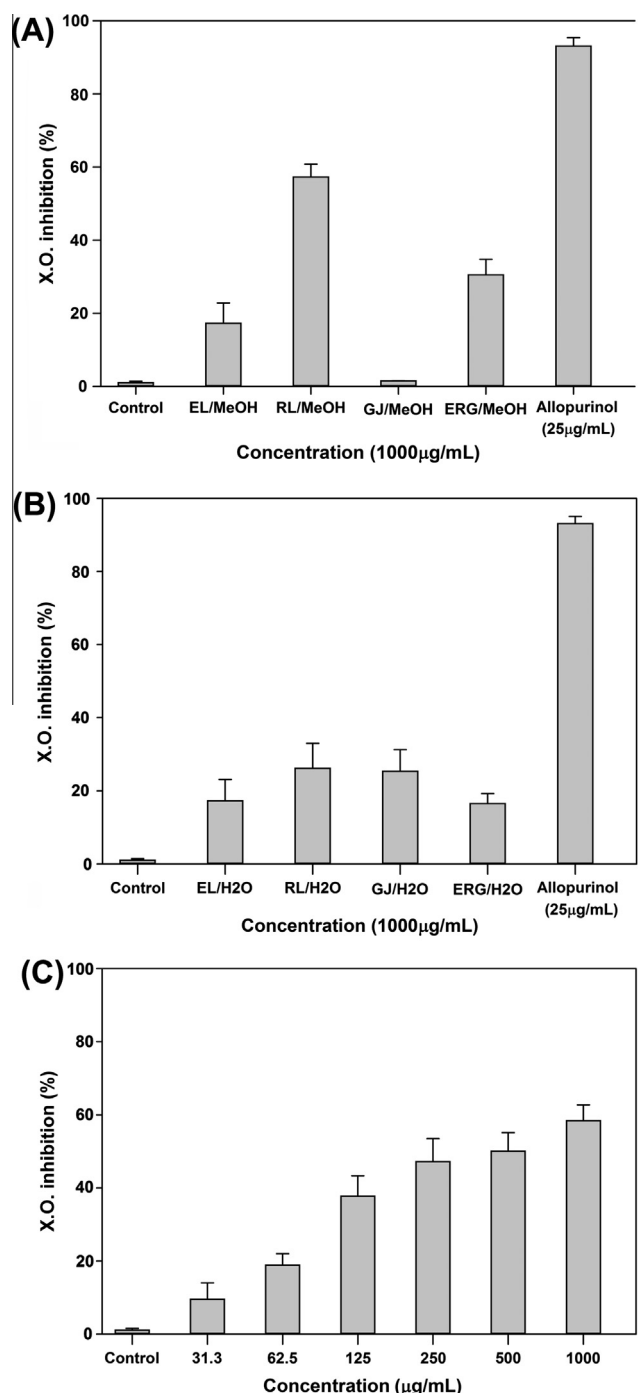


Fig. 1. Xanthine oxidase inhibitory activity of methanol and water extracts isolated from the EL, RL, GJ and ERG *in vitro*. Inhibitory effects by (A) 1000 µg/mL EL/MeOH, 1000 µg/mL RL/MeOH, 1000 µg/mL GJ/MeOH, 1000 µg/mL ERG/MeOH, and 25 µg/mL allopurinol, (B) 1000 µg/mL EL/H₂O, 1000 µg/mL RL/H₂O, 1000 µg/mL GJ/H₂O, 1000 µg/mL ERG/H₂O and 25 µg/mL allopurinol, (C) various concentration of RL/MeOH on XO activity. Each value is represented as mean ± S.D. of three separate experiments in triplicate measurements.

therefore, we next investigated its effects *in vivo*. Administration of potassium oxonate to rats (250 mg/kg body weight, i.p.) after 180 min resulted in a significant increase in the serum urate level when compared to normal control animals. Animals treated with 0.13 and 0.39 g/kg ERG/MeOH after 180 min showed a significant decrease in the serum urate level up to 1.92 ± 0.68 mg/dL and 1.68 ± 0.37 mg/dL, respectively, when compared to hyperuricemic control. The effect of allopurinol treated group is almost similar to the serum urate level of normal control group (Table 3). Fig. 2

shows the effect of hypouricaemic activity after treatment with ERG/MeOH at 0, 60, 90, 120, 180, 270 min, respectively. The study indicated that the level of serum uric acid was decreased after oral administration of ERG/MeOH. In Fig. 2 (Table 3), one-way ANOVA for each time point are 60 min: $F(5,42) = 51.574$, $P < 0.001$; 90 min: $F(5,42) = 85.560$, $P < 0.001$; 120 min: $F(5,42) = 104.119$, $P < 0.001$; 180 min: $F(5,42) = 66.995$, $P < 0.001$; 270 min: $F(5,42) = 20.269$, $P < 0.001$; $n = 8$ in each group.

3.6. Acetic acid-induced writhing

The treatment with ERG/MeOH (0.13 g/kg and 0.39 g/kg) reduced the abdominal writhes from 24.50 ± 9.81 (control group) to 14.67 ± 4.50 and $6.30 \pm 4.95\%$, respectively (Fig. 3). The study of acetic acid-induced writhing was reanalyzed using one-way ANOVA, and its F value is $F(3,30) = 23.180$, $P < 0.001$ ($n = 8-10$).

3.7. Formalin-induced nociception

The treatment with ERG/MeOH (0.13 g/kg and 0.39 g/kg) did not reduced the licking time (time licking the paw after intraplantar injection of formalin) at the neurogenic phase, but reduced the licking time (sec) at the inflammatory phase from 780.08 ± 44.02 s (control group) to 454.83 ± 25.56 s and 243.75 ± 45.43 s. Indomethacin reduced the licking time only at the second phase (Fig. 4). One-way ANOVA was used to analyze the data in the study of formalin-induced nociception at phase 2, and its F value is $F(3,25) = 231.578$, $P < 0.001$ ($n = 7-8$).

3.8. Acute toxicity studies

In LD₅₀ studies, it was found that the animals were safe up to a maximum dose of 5 g/kg body weight. There were no changes and normal behavior pattern and no signs and symptoms of toxicity and mortality were observed.

4. Discussion

The decoction of ERG is a traditionally recognized folk medicine used to treat gout in Taiwan. In the course of investigating the therapeutic effects of dietary herbal teas on gout, we developed a renewed interest in the XO-inhibitory ability of methanol and aqueous extracts from the EL, RL, GJ and ERG. The present study demonstrated that variation in methods of extraction procedures could result in a different pattern of active compounds and different magnitude of pharmacological activities. In the study, the results suggested that methanol extracts were better than water extracts for inhibition of XO activity and DPPH radical scavenging activity, except the water extract of GJ, which exhibited the strongest radical scavenging effect.

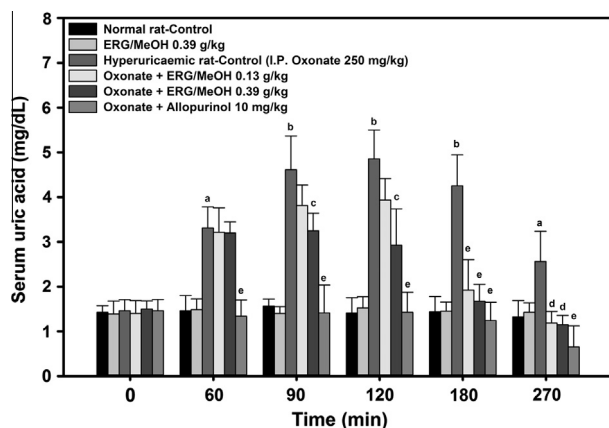
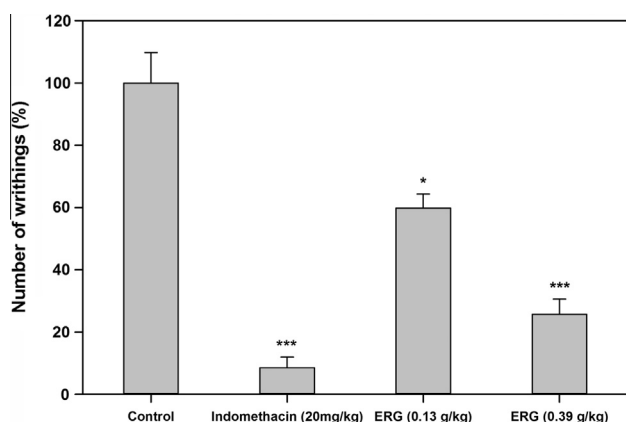
Our results showed that total phenolic compounds were enriched in the *G. jasminoides*. Polyphenolic compounds have been shown to possess antioxidant and anti-inflammatory properties (Middleton et al., 2000). Variations in antioxidant capacity of methanol and water extracts may be attributed to differences in their phenolic contents. All the extracts showed an increase in antioxidant capacity with increase in polyphenolic content. The antioxidant activity shown by the EL, RL, and GJ extracts maybe due to the presence of polyphenols, such as gallic acid and chlorogenic acid.

A number of anti-hyperuricemic agents are available; their utilization is sometimes limited by the associated undesirable side effects. Many studies have shown a growing interest for the protective role of XO inhibitor allopurinol on uric acid levels of hyperuricemia and gout. However, about 5% of patients are unable to

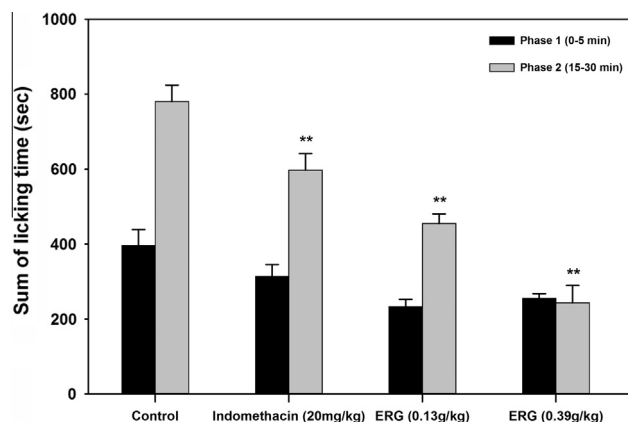
Table 3

Effect of the plant extracts on serum urate level in oxonate-induced hyperuricaemia in rat at 180 min.

Treatment groups	Dosage of drugs (mg/kg)	Serum urate levels (mg/dl)
Normal rat-control	–	1.44 ± 0.34
Normal rat dosed with ERG/MeOH	390	1.45 ± 0.21
Hyperuricemic rat-control	–	4.25 ± 0.69 ^a
Hyperuricemic rat dosed with ERG/MeOH	130	1.92 ± 0.68 ^b
Hyperuricemic rat dosed with ERG/MeOH	390	1.68 ± 0.37 ^b
Hyperuricemic rat dosed with allopurinol	10	1.24 ± 0.41 ^b

The hyperuricemic rat were produced by potassium oxonate (250 mg/kg) pretreatment. Values are mean ± S.D., *n* = 8 in each group.^a *P* < 0.001 significant when compared to normal control.^b *P* < 0.001 significant when compared to hyperuricemic control.**Fig. 2.** Hypouricaemic activity of the methanol extract of ERG in hyperuricemic rat model. Rats pretreated with potassium oxonate (250 mg/kg, i.p.) and then treatment with 0.13 g/kg and 0.39 g/kg ERG/MeOH and 10 mg/kg allopurinol at 0, 60, 90, 120, 180, 270 min, respectively. The serum urate level was determined. Values are mean ± S.D., *n* = 8 in each group. ^a*P* < 0.01 and ^b*P* < 0.001, when compared to normal control; ^c*P* < 0.05, ^d*P* < 0.01 and ^e*P* < 0.001 significant when compared to hyperuricemic control.**Fig. 3.** Effect of ERG/MeOH (0.13 g/kg and 0.39 g/kg, p.o.) in the number of acetic acid induced writhing in mice. Indomethacin (20 mg/kg, p.o.) was used as positive control. Vertical bars represent mean ± S.D. in percentage of control group of accumulated writhes in 30 min for each experimental group. ^{*}*P* < 0.05; ^{**}*P* < 0.01; ^{***}*P* < 0.001 when compared to control group.

tolerate its adverse side effects, which include gastrointestinal irritation, bone marrow suppression, and hypersensitivity syndromes ranging from simple skin rash to life-threatening conditions in which the patients develop toxic epidermal necrolysis, fever, hepatitis, eosinophilia, and worsening renal function (Fathallah et al., 2010; Fritsch and Sidoroff, 2000; Horiuchi et al., 2000). Uricosuric agents are used in patients with allopurinol-allergic syndromes as well as in underexcretors with normal renal function and no

**Fig. 4.** Effect of ERG/MeOH (0.13 g/kg and 0.39 g/kg, p.o.) in formalin induced pain in mice on first phase (0–5 min) and second phase (15–30 min). Indomethacin (20 mg/kg, p.o.) were used as positive control. Vertical bars represent mean ± S.D. of licking time, in seconds. ^{**}*P* < 0.001 compared with the control group of the second phase.

history of urolithiasis. These uricosuric agents are also not without problems. For example, benzbromarone can cause fulminant hepatotoxicity, and this in fact had led to its withdrawal from the French market in 2001 (Bieber and Terkeltaub, 2004), whereas probenecid and sulfinpyrazone have been reported to be nephrotoxic when used to treat hyperuricemia associated with moderate chronic renal insufficiency (Neogi, 2011).

Therefore, the search for better anti-hyperuricemic agents is highly warranted and our study provides a better option for treatment of gout. From our studies on folk medicine, *R. lanceolata* emerged as a potent uric acid-lowering agent. *R. lanceolata* is an important Chinese folk medicine which possesses circulation-promoting, expectorant, against cough properties and antitumor activities. In our earlier study, we have identified the constituents of *R. lanceolata* as mollugin, 1-hydroxy-2-methyl-9,10-anthraquinone, rubiadin, alizarin, lucidin ethyl ether, 2-methylquinizarin, 3-carbomethoxy-1-hydroxy-9,10-anthraquinone, and ω-hydroxypachybasin (Kuo et al., 1995).

Our results suggested that the water extract of GJ exhibited the strongest scavenging effect on DPPH radical. It has been suggested that some active compounds in the aqueous extracts of GJ were responsible for its XO inhibitory activity. Three active compounds were isolated from the aqueous extracts of GJ, namely geniposide, crocin-1, and geriposidic acid. Recently, crocins and geniposide, the compounds responsible for the pharmacological properties of GJ, have been reported to contribute significantly to the radical scavenging properties of this drug (Chen et al., 2010). In the future, we will test whether these active compounds have beneficial effects on XO inhibitory activity.

In Taiwan, the indigenous people extract 37.5 g dried ERG and take thrice daily for the treatment of gout. Due to the yield of

extract is 3.5%, therefore, the oral dose of human is 3.9375 g/day for a 60 kg human (65.625 mg/kg/day). The coefficient of human to rat is about 6.25, therefore, the oral dose for rat is 393.75 mg/kg/day. For this reason, rats were treated with 130 mg/kg and 390 mg/kg ERG/MeOH in hypouricaemic activity study.

The acetic acid-induced abdominal writhing test is a high sensitivity but low specificity model (Koster et al., 1959). It was demonstrated that ERG/MeOH possessed antinociceptive activity. The formalin-induced pain test is divided in two phases: the neurogenic phase (first 5 min after formalin intraplantar injection) that involves a direct activation of sensory fibers C by formaldehyde as well as participation of peripheral mediators stimulating directly the nociceptors; and inflammatory phase (15–30 min after the injection) that involves the release and production of inflammatory mediators (Rocha et al., 2007). In addition, we investigated the inhibitory effects of ERG/MeOH on the induction of NO synthase (NOS) and COX-2 in RAW 264.7 cells activated with lipopolysaccharide (LPS). Western blotting demonstrated that ERG/MeOH significantly blocked protein expression of iNOS and COX-2 in LPS-induced macrophages (Supplementary Fig. 1). The ERG/MeOH showed an anti-nociceptive activity only in the second phase, and possibly due to an anti-inflammatory action, such as inhibition in synthesis or release of inflammation mediators (prostaglandins, thromboxanes, leukotrienes, among others).

5. Conclusions

In the present study, three plants from Taiwan were selected based on their ethnomedical use for the treatment of gout by the natives of this region. Our study revealed that the aqueous extract of GJ was one of the positive samples to have a potent XO inhibitory effect in comparison to allopurinol. This study presents data to explain the effect of ERG/MeOH in treating hyperuricemia and gout and it has the potential to help thousands of patients who suffer from gout in Taiwan and around the world.

Conflict of Interest

The authors declare that there are no conflicts of interest.

Transparency Document

The Transparency document associated with this article can be found in the online version.

Acknowledgment

D.-H. Kuo, L.-J. Huang, and T.-D. Way contributed equally to this work.

Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.fct.2014.05.004>.

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